

EXERCISE SHEET 7

INTEGRATION IN VECTOR FIELDS

① Evaluate the line integral where C is the given curve

Ⓐ $\int_C y \, ds$ $C: x=t^2 \quad y=t \quad 0 \leq t \leq 2$

Ⓑ $\int_C y/x \, ds$ $C: x=t^4 \quad y=t^3 \quad 1/2 \leq t \leq 1$

Ⓒ $\int_C xy^4 \, ds$ $C: \text{right half of the circle } x^2+y^2=16$

Ⓓ $\int_C ye^x \, ds$ $C: \text{the line segment joining } (1,2) \text{ to } (4,7)$

Ⓔ $\int_C (xy+lx) \, dy$ $C: \text{arc of the parabola } y=x^2 \text{ from } (1,1) \text{ to } (3,9)$

Ⓕ $\int_C x e^y \, dx$ $C: x=e^y \text{ from } (1,0) \text{ to } (e,1)$

Ⓖ $r(t) = \langle t^2, t \rangle$ $f(x,y) = y \quad f(r(t), t) = t$
 $r'(t) = \langle 2t, 1 \rangle \quad |r'(t)| = \sqrt{4t^2+1}$

$$\int_0^2 t \sqrt{4t^2+1} \, dt \stackrel{u}{=} \frac{1}{8} \cdot \frac{(4t^2+1)^{3/2}}{3/2} \bigg|_0^2 = \frac{1}{12} (17\sqrt{17}-1)$$

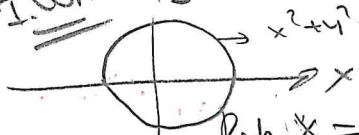
Ⓑ $r(t) = \langle t^4, t^3 \rangle \quad r'(t) = \langle 4t^3, 3t^2 \rangle \quad |r'(t)| = \sqrt{16t^6+9t^4}$
 $f(x,y) = y/x \quad f(t^4, t^3) = \frac{t^3}{t^4} = \frac{1}{t}$

$$\int_C y/x \, ds = \int_{1/2}^1 \frac{1}{t} \sqrt{16t^6+9t^4} \, dt = \int_{1/2}^1 \frac{1}{t} \cdot t^2 \sqrt{16t^2+9} \, dt$$

$$= \frac{(16t^2+9)^{3/2}}{3/2} \cdot \frac{1}{32} \bigg|_{1/2}^1 = \frac{1}{48} (25^{3/2} - 13^{3/2})$$

Darya Altintop

Ⓒ I want xy



Right: $x = \sqrt{16-y^2}$

①

$y=t \quad x = \sqrt{16-t^2}$
 $-4 \leq t \leq 4$

$r(t) = \langle \sqrt{16-t^2}, t \rangle \quad r'(t) = \langle \frac{-t}{\sqrt{16-t^2}}, 1 \rangle$

$|r'(t)| = \sqrt{\frac{t^2}{16-t^2} + 1} = \frac{4}{\sqrt{16-t^2}}$

$\int_C xy^4 \, ds = \int_{-4}^4 \frac{t^2}{\sqrt{16-t^2}} \cdot t^4 \cdot \frac{4}{\sqrt{16-t^2}} \, dt = 4 \int_{-4}^4 \frac{t^6}{16-t^2} \, dt = 4 \cdot 2 \cdot 45 = 240$

II. WAY $x = 4 \cos t$ $y = 4 \sin t$ $-\pi/2 \leq t \leq \pi/2$

$$\int_C xy^4 ds = \int_{-\pi/2}^{\pi/2} (4 \cos t)(4 \sin t)^4 \sqrt{(-4 \sin t)^2 + (4 \cos t)^2} dt$$

$$= \int_{-\pi/2}^{\pi/2} 4^5 \cos t \sin^4 t \sqrt{16(\sin^2 t + \cos^2 t)} dt$$

$$= 4^5 \int_{-\pi/2}^{\pi/2} \sin^4 t \cdot \cos t \cdot 4 dt = 4^6 \cdot \left[\frac{\sin^5 t}{5} \right]_{-\pi/2}^{\pi/2} = \frac{2 \cdot 4^6}{5}$$

(d) $(x_0, y_0) = (1, 2)$ $x(t) = (1-t)x_0 + t \cdot x_1 = (1-t) \cdot 1 + 4t = 1+3t$
 $(x_1, y_1) = (4, 7)$ $y(t) = (1-t)y_0 + t \cdot y_1 = (1-t) \cdot 2 + 7t = 2+5t$
 $r(t) = \langle x(t), y(t) \rangle$ $r'(t) = \langle 3, 5 \rangle$ $0 \leq t \leq 1$
 $|r'(t)| = \sqrt{34}$

$$\int_C ye^x ds = \int_0^1 (2+5t) e^{1+3t} \sqrt{34} dt = \sqrt{34} e \int_0^1 (2+5t) \cdot e^{3t} dt$$

$$= \sqrt{34} e \left[(2+5t) \frac{e^{3t}}{3} - \int \frac{5e^{3t}}{3} dt \right]$$

$$= \sqrt{34} e \left[(2+5t) \frac{e^{3t}}{3} - \frac{5}{9} e^{3t} \right]_0^1$$

$$= \frac{\sqrt{34}}{9} (16e^4 - e)$$

$2+5t = u$ $5dt = du$
 $e^{3t} dt = dv$ $v = \frac{e^{3t}}{3}$

(e) $y = x^2$ from $(1,1)$ to $(3,9)$, $x=t$ $y=t^2$ $1 \leq t \leq 3$
 $dy = 2t dt$

$$\int_C (xy + \ln x) dy = \int_1^3 (t \cdot t^2 + \ln t) \cdot 2t dt$$

$$= 2 \int_1^3 (t^4 + t \ln t) dt = 2 \left[\frac{t^5}{5} + \ln t \cdot \frac{t^2}{2} - \frac{t^4}{4} \right]_1^3 = \frac{464}{5} + 9 \ln 3$$

$$\int t \ln t dt = \ln t \cdot \frac{t^2}{2} - \int \frac{1}{t} \cdot \frac{t^2}{2} dt = \ln t \cdot \frac{t^2}{2} - \frac{t^4}{4}$$

$$\ln t = u \quad \frac{1}{t} dt = du$$

$$t dt = dv \quad v = \frac{t^2}{2}$$

Peyar Attender

① $x = e^t$ $y = t \Rightarrow x = e^t \quad r(t) = \langle e^t, t \rangle$

$(1, 0)$ $(e, 1)$ $dx = e^t dt$
 y changes from 0 to 1 so $0 \leq t \leq 1$

$$\int_C x \cdot e^{xy} dx = \int_0^1 e^t \cdot e^t \cdot e^t dt = \int_0^1 e^{3t} dt = \frac{e^{3t}}{3} \Big|_0^1 = \frac{e^3 - 1}{3}$$

② $\int_C \sin x dx + \cos y dy$ Consists of the top half of the circle

$x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$ and the line segment from $(-1, 0)$ to $(-2, 3)$

$C = C_1 + C_2$

C_1 : $x = \cos t$ $dx = -\sin t dt$
 $y = \sin t$ $dy = \cos t dt$
 $0 \leq t \leq \pi$

Use $x(t) = (1-t)x_0 + t x_1$
 $y(t) = (1-t)y_0 + t y_1$
 C_2 : $x = -1-t$ $dx = -dt$
 $y = 3t$ $dy = 3dt$
 $0 \leq t \leq 1$

Then: $\int_C \sin x dx + \cos y dy = \int_{C_1} \sin x dx + \cos y dy + \int_{C_2} \sin x dx + \cos y dy$

$$= \int_0^\pi \sin(\cos t) (-\sin t) dt + \cos(\sin t) \cos t dt$$

$$+ \int_0^1 \sin(-1-t) (-dt) + \cos(3t) \cdot 3 dt$$

$$= \left[\cos(\cos t) + \sin(\sin t) \right]_0^\pi + \left[-\cos(-1-t) + \sin 3t \right]_0^1$$

$$= -\cos(\cos \pi) + \sin(\sin \pi) + \cos(\cos 0) - \sin(\sin 0)$$

$$- \cos(-2) + \sin(3) + \cos(-1) - \sin(0) = \cos 1 - \cos 2 + \sin 3$$

③ $\int_C x e^{y^2 z} dz$

C is the segment from $(0, 0, 0)$ to $(1, 2, 3)$

$x(t) = x_0(1-t) + x_1 t = t$ $dx = dt$
 $y(t) = y_0(1-t) + y_1 t = 2t$ $dy = 2dt$
 $z(t) = z_0(1-t) + z_1 t = 3t$ $dz = 3dt$
 $0 \leq t \leq 1$

$r(t) = \langle x(t), y(t), z(t) \rangle$ $r'(t) = \langle 1, 2, 3 \rangle$

$|r'(t)| = \sqrt{1+4+9} = \sqrt{14}$

$$\int_0^1 t \cdot e^{2t \cdot 3t} \sqrt{14} dt$$

$$= \frac{e^{6t^2}}{12} \sqrt{14} \Big|_0^1 = \frac{\sqrt{14}}{12} (e^6 - 1)$$

③

Deyan Hristov

4) $\int_C z dx + x dy + y dz = ?$ where $x=t^2$ $y=t^3$ $z=t^2$ $0 \leq t \leq 1$

$dx = 2t dt$ $dy = 3t^2 dt$ $dz = 2t dt$

$\int_0^1 t^2 \cdot 2t dt + t^2 \cdot 3t^2 dt + t^3 \cdot 2t dt$

$= \int_0^1 (2t^3 + 3t^4 + 2t^4) dt = \left[\frac{2t^4}{4} + \frac{5t^5}{5} \right]_0^1 = \frac{2}{4} + 1 = 1 + \frac{1}{2} = \frac{3}{2}$

5) $\int_C (x+yz) dx + 2x dy + xyz dz = ?$

where C consists of the line segment from $(1,0,1)$ to $(2,3,1)$ and from $(2,3,1)$ to $(2,5,2)$

C_1 : from $(1,0,1)$ to $(2,3,1)$ $x(t) = (1-t) \cdot 1 + 2t = t+1$ $dx = dt$
 $y(t) = 3t$ $dy = 3dt$ $z(t) = (1-t) \cdot 1 + t \cdot 1 = 1$ $dz = 0$

C_2 : from $(2,3,1)$ to $(2,5,2)$ $x(t) = (1-t) \cdot 2 + t \cdot 2 = 2$ $dx = 0$
 $y(t) = (1-t) \cdot 3 + t \cdot 5 = 3+2t$ $dy = 2dt$
 $z(t) = (1-t) \cdot 1 + t \cdot 2 = t+1$ $dz = dt$

$\int_0^1 ((t+1) + 3t \cdot 1) \cdot dt + 2 \cdot (t+1) \cdot 3dt + (t+1)(3t) \cdot 1 \cdot dt$
 $+ \int_0^1 (1 + (1+2t)(t+1)) \cdot 0 + 2 \cdot 2 \cdot 2dt + 2 \cdot (3+2t)(t+1) \cdot dt$
 $= \int_0^1 (x+yz) dx + 2x dy + xyz dz$

$= \int_0^1 (10t+7) dt + \int_0^1 (4t^2+10t+14) dt =$

$= (5t^2+7t) \Big|_0^1 + \left(\frac{4}{3}t^3 + 5t^2 + 14t \right) \Big|_0^1 = \frac{97}{3}$

(4)

Deyan Atanasov

6) Evaluate the line integral $\int_C F \cdot dr$ where C is given by the vector function $r(t)$

(a) $F(x,y) = x^2 y^3 \vec{i} - y \sqrt{x} \vec{j}$ $r(t) = t^2 \vec{i} - t^3 \vec{j}$ $0 \leq t \leq 1$

(b) $F(x,y,z) = yz \vec{i} + xz \vec{j} + xy \vec{k}$ $r(t) = t \vec{i} + t^2 \vec{j} + t^3 \vec{k}$ $0 \leq t \leq 2$

(c) $F(x,y,z) = z \vec{i} + y \vec{j} - x \vec{k}$ $r(t) = t \vec{i} + \sin t \vec{j} + \cos t \vec{k}$ $0 \leq t \leq \pi$

(d) $F(x,y) = e^{x-1} \vec{i} + xy \vec{j}$ $r(t) = t^2 \vec{i} + t^3 \vec{j}$ $0 \leq t \leq 1$

(e) $F(x,y,z) = x \vec{i} - z \vec{j} + y \vec{k}$ $r(t) = 2t \vec{i} + 3t \vec{j} - t^2 \vec{k}$ $-1 \leq t \leq 1$

(a) $F(r(t)) = (t^2)^2 (-t^3)^3 \vec{i} - (-t^3) \sqrt{t^2} \vec{j} = -t^3 \vec{i} + t^4 \vec{j}$
 $r'(t) = 2t \vec{i} - 3t^2 \vec{j}$

$$\int_C F \cdot dr = \int_0^1 F(r(t)) \cdot r'(t) dt = \int_0^1 (-12t^{14} - 3t^6) dt = \left(-\frac{12}{15} t^{15} - \frac{3}{7} t^7 \right) \Big|_0^1 = -\frac{12}{15} - \frac{3}{7}$$

(b) $F(r(t)) = (t^2)(t^3) \vec{i} + t \cdot t^3 \vec{j} + t \cdot t^2 \vec{k}$
 $r(t) = \vec{i} + 2t \vec{j} + 3t^2 \vec{k}$ $0 \leq t \leq 2$

$$\int_C F \cdot dr = \int_0^2 F(r(t)) \cdot r'(t) dt = \int_0^2 (t^5 + 2t^5 + 3t^5) dt = t^6 \Big|_0^2 = 64$$

(c) $F(r(t)) = \cos t \vec{i} + \sin t \vec{j} - t \vec{k}$
 $r'(t) = \vec{i} + \cos t \vec{j} - \sin t \vec{k}$

$$\begin{aligned} \int_C F \cdot dr &= \int_0^\pi \langle \cos t, \sin t, -t \rangle \cdot \langle 1, \cos t, -\sin t \rangle dt = \int_0^\pi (\cos t + \sin t \cos t - t \sin t) dt \\ &= \left[\sin t + \frac{\sin^2 t}{2} + (t \cos t - \sin t) \right]_0^\pi = \pi \end{aligned}$$

(d) $F(r(t)) = e^{t^2-1} \vec{i} + t^5 \vec{j}$
 $r'(t) = \langle 2t, 3t^2 \rangle$

Danya Altintan

$$\begin{aligned} \int_C F(r(t)) \cdot r'(t) dt &= \int_0^1 \langle e^{t^2-1}, t^5 \rangle \cdot \langle 2t, 3t^2 \rangle dt = \int_0^1 e^{t^2-1} 2t dt \\ &+ \int_0^1 3t^7 dt = \left(\frac{e^{t^2-1}}{5} \right) \Big|_0^1 + \frac{3 \cdot t^8}{8} \Big|_0^1 = \frac{11}{8} - \frac{1}{5} e \end{aligned}$$

② $F(r(t)) = 2t\vec{i} + t^2\vec{j} + 3t\vec{k}$
 $r'(t) = 2\vec{i} + 3\vec{j} - 2t\vec{k}$

$f(x) = f(x^{1/2})$

$$\int_{-1}^1 F(r(t)) \cdot r'(t) dt = \int_{-1}^1 \langle 2t, t^2, 3t \rangle \cdot \langle 2, 3, -2 \rangle dt$$

$$= \int_{-1}^1 (4t + 3t^2 - 6t) dt = \int_{-1}^1 (3t^2 - 2t) dt = \left. t^3 - t^2 \right|_{-1}^1 = (1-1) - (-1-1) = -2$$

⑦ Find the work done for the force field along the given curves

⑧ $F(x,y) = x\vec{i} + (y+2)\vec{j}$ $r(t) = (t - \sin t)\vec{i} + (1 - \cos t)\vec{j}$ $0 \leq t \leq 2\pi$

⑨ $F(x,y) = x \sin y \vec{i} + y \vec{j}$ $y = x^2$ from $(-1,1)$ to $(2,4)$

⑩ $F(x,y,z) = \langle y+z, x+z, x+y \rangle$ a line segment from $(1,0,0)$ to $(3,4,2)$

⑪ $\int_C F \cdot dr = \int_0^{2\pi} \langle t - \sin t, 1 - \cos t \rangle \cdot \langle 1 - \cos t, \sin t \rangle dt$

$r(t) = \langle 1 - \cos t, \sin t \rangle$

$$= \int_0^{2\pi} (t - t \cos t - \sin t + \sin t \cos t + 3 \sin t - \sin t \cos t) dt$$

$$= \int_0^{2\pi} (t - t \cos t + 2 \sin t) dt$$

$$= \left. \frac{t^2}{2} - (t \sin t + \cos t) - 2 \cos t \right|_0^{2\pi} = 2\pi^2$$

$$\int t \cos t dt = t \cdot \sin t - \int \sin t dt = t \cdot \sin t + \cos t$$

$t = u$ $\cos t dt = du$

$dt = du$ $\sin t = u$

Doyen Altıntaş

(b) $x=t \quad y=t^2 \quad -1 \leq t \leq 2 \quad r(t) = \langle t, t^2 \rangle \quad r'(t) = \langle 1, 2t \rangle$

$$W = \int_{-1}^2 \langle t \cos t^2, t^2 \rangle \cdot \langle 1, 2t \rangle dt$$

$$F(r(t)) = \langle t \cos t^2, t^2 \rangle$$

$$= \int_{-1}^2 (t \cos(t^2) + 2t^3) dt = -\frac{\cos t^2}{2} + \frac{2t^4}{4} \Big|_{-1}^2$$

$$= \frac{1}{2}(15 + \cos 1 - \cos 4)$$

(c) $x(t) = (1-t) \cdot 1 + 3t = 1+2t \quad W = \int F(r(t)) \cdot r'(t) dt$

$$y(t) = (1-t) \cdot 0 + 4t = 4t$$

$$z(t) = (1-t) \cdot 0 + 2t = 2t$$

$$r(t) = \langle 1+2t, 4t, 2t \rangle \quad 0 \leq t \leq 1$$

$$r'(t) = \langle 2, 4, 2 \rangle$$

$$= \int_0^1 \langle 6t, 1+4t, 1+6t \rangle \cdot \langle 2, 4, 2 \rangle dt$$

$$= \int_0^1 (12t + 16t + 4 + 2 + 12t) dt$$

$$= 20t^2 + 6t \Big|_0^1 = 26$$

$$F(r(t)) = \langle 4t + 2t, 1+2t + 2t, 1+2t + 4t \rangle$$

$$= \langle 6t, 1+4t, 1+6t \rangle$$

(8) Determine whether or not F is conservative vector field.

If it is find f

(a) $F(x,y) = (6x+5y)\vec{i} + (5x+4y)\vec{j}$

(b) $F(x,y) = (x^3 + 4xy)\vec{i} + (4xy - y^3)\vec{j}$

(c) $F(x,y) = xe^{4y}\vec{i} + ye^{x^3}\vec{j}$

(d) $F(x,y) = (2x \cos y - y \cos x)\vec{i} + (-x^2 \sin y - \sin x)\vec{j}$

(e) $F(x,y) = P(x,y)\vec{i} + Q(x,y)\vec{j}$

Check $P_y = Q_x = ?$

$$P = 6x+5y \quad Q = 5x+4y$$

$$P_y = 5 = Q_x = 5 \quad \checkmark$$

Conservative

Find f such that $F = \nabla f$

$$\langle P, Q \rangle = \langle f_x, f_y \rangle$$

$$\langle 6x+5y, 5x+4y \rangle = \langle f_x, f_y \rangle$$

$$f_x = 6x+5y$$

$$f_y = 5x+4y$$

(7)

Deyan Altunlar

$$f(x,y) = 3x^2 + 5xy + h(y)$$

$$\frac{\partial f}{\partial y} = 5x + h'(y) = 5x + 4y$$

$$h'(y) = 4y \quad h(y) = 2y^2 + C$$

$$f(x,y) = 3x^2 + 5xy + 2y^2 + C$$

(b) $P(x,y) = x^3 + 4xy$

$$P_y = Q_x$$

$$4x \neq 4y$$

NOT CONSERVATIVE

$$Q(x,y) = 4xy - y^3$$

NO potential function.

(c) $P(x,y) = xe^y$

$$P_y = Q_x ?$$

$$Q(x,y) = ye^x$$

$$P_y = xe^y$$

$$Q_x = ye^x$$

$$P_y \neq Q_x$$

NOT CONSERVATIVE

NO POTENTIAL

(d)

$$P(x,y) = 2x \cos y - y \cos x$$

$$P_y = -2x \sin y - \cos x$$

$$Q(x,y) = -x^2 \sin y - \sin x$$

$$Q_x = -2x \sin y - \cos x$$

$P_y = Q_x$ F is conservative. There is a potential function

f such that $F = \nabla f$

Find f .

$$f_x = 2x \cos y - y \cos x$$

$$f(x,y) = x^2 \cos y - y \sin x + g(y)$$

$$f_y = -x^2 \sin y - \sin x$$

$$\frac{\partial f}{\partial y} = -x^2 \sin y - \sin x + g'(y) = -x^2 \sin y - \sin x$$

$$g'(y) = 0 \quad g(y) = C$$

$$f = x^2 \cos y - y \sin x + C$$

(9) For the following questions.

(a) Find the function f such that $F = \nabla f$

(b) Use (a) to evaluate $\int_C F \cdot dr$.

Deja Altimeter

(2,1)

(a) $F(x,y) = y\vec{i} + (x+2y)\vec{j}$ C is the upper semicircle starts at (0,1) and ends at

(b) $F(x,y) = x^3 y^4 \vec{i} + x^4 y^3 \vec{j}$ $r(t) = \sqrt{t} \vec{i} + (1+t^3) \vec{j} \quad 0 \leq t \leq 1$

(c) $F(x,y) = \frac{y^2}{1+x^2} \vec{i} + 2y \arctan x \vec{j}$ $r(t) = t^2 \vec{i} + 2t \vec{j} \quad 0 \leq t \leq 1$

(8)

a) ① $f_x(x,y) = y \Rightarrow f(x,y) = xy + h(y)$
 $f_y(x,y) = x + 2y \quad f_y = x + h'(y) = x + 2y$
 $h'(y) = 2y \quad h(y) = y^2 + C$

$f(x,y) = xy + y^2 + C$

② $\int_C F \cdot dr = \int \nabla f \cdot dr = f(B) - f(A)$

C is a curve containing A and B

$\int_C F \cdot dr = f(2,1) - f(0,1) = (2 + 1 + C) - (1 + C) = \underline{2}$

b) ① $f_x(x,y) = x^3 y^4 \quad f(x,y) = \frac{x^4}{4} y^4 + h(y)$
 $f_y(x,y) = x^4 y^3 \quad f_y(x,y) = x^4 y^3 + h'(y) = x^4 y^3$
 $h'(y) = 0 \quad h(y) = C$

$f(x,y) = \frac{x^4}{4} y^4 + C$

② Partial $r(0) = 0\vec{i} + \vec{j} = (0,1)$
 final $r(1) = \vec{i} + 2\vec{j} = (1,2)$

$\int_C F \cdot dr = \int \nabla f \cdot dr = f(1,2) - f(0,1) = \underline{4}$

c) ① $f_x = \frac{y^2}{1+x^2} \quad f_y = 2y \arctan x$
 $\Downarrow$

$f(x,y) = \int f_x dx = \arctan x \cdot y^2 + h(y)$

$f_y = 2y \arctan x + h'(y) = 2y \arctan x$
 $h'(y) = 0 \quad h(y) = C$

Derive Arctan

$f(x,y) = y^2 \arctan x + C$

② Partial point $r(0) = 0\vec{i} + 0\vec{j} = (0,0)$
 End point $r(1) = \vec{i} + 2\vec{j} = (1,2)$

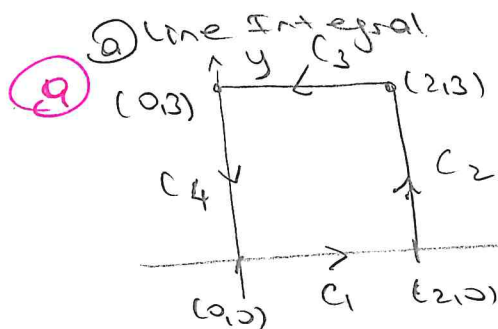
$\int_C F \cdot dr = \int \nabla f \cdot dr = f(r(1)) - f(r(0))$
 $= 4 \arctan 1$
 $= 4 \cdot \pi/4 = \pi$

10) Evaluate the following integrals by two methods

a) Line integral b) Green's Theorem.

a) $\int_C xy^2 dx + x^3 dy$ C is the rectepde with vertices (0,0), (2,0), (2,3) and (0,3)

b) $\int_C x dx + y dy$ C: line segments from (0,1) to (0,0), from (0,0) to (1,0) and the parabola $y=1-x^2$ from (1,0) to (0,1)



$C_1: x=t \quad dx=dt \quad y=0 \quad dy=0 \quad 0 \leq t \leq 2$
 $C_2: x=2 \quad dx=0 \quad y=t \quad dy=dt \quad 0 \leq t \leq 3$
 $C_3: x=2-t \quad dx=-dt \quad y=3 \quad dy=0 \quad 0 \leq t \leq 2$
 $C_4: x=0 \quad dx=0 \quad y=3-t \quad dy=-dt \quad 0 \leq t \leq 3$

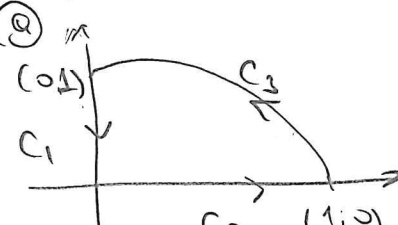
$$\int_C xy^2 dx + x^3 dy = \sum_{i=1}^4 \int_{C_i} xy^2 dx + x^3 dy$$

$$\begin{aligned} \int_C xy^2 dx + x^3 dy &= \underbrace{\int_0^2 0 dt}_{C_1} + \underbrace{\int_0^3 8 dt}_{C_2} + \underbrace{\int_0^2 -9(2-t) dt}_{C_3} + \underbrace{\int_0^3 0 dt}_{C_4} \\ &= 24 - 18 = 6 \end{aligned}$$

$Qx = 3x^2 \quad Py = 2xy$

b) $\int_0^3 \int_0^2 (Qx - Py) dx dy = \int_0^3 \int_0^2 (3x^2 - 2xy) dx dy$
 $= \int_0^3 (x^3 - x^2 y) \Big|_0^2 dy = \int_0^3 (8 - 4y) dy = \left[4y - 2y^2 \right]_0^3 = 24 - 18 = 6$

Daya Attintan

(b) (a) 

$C_1: x=0 \quad y=1-t \quad dx=0 \quad dy=-dt \quad 0 \leq t \leq 1$

$$\int_{C_1} x dx + y dy = \int_0^1 (1-t)(-dt) = -\left(t - \frac{t^2}{2}\right) \Big|_0^1 = -1/2$$

$C_2: y=0 \quad x=t \quad 0 \leq t \leq 1 \quad dx=dt \quad dy=0$

$$\int_{C_2} x dx + y dy = \int_0^1 t dt = \frac{1}{2}$$

$C_3: y=1-x^2 \quad x=t \quad y=1-t^2 \quad -1 \leq t \leq 0$

yo da (II) $x=1-t \quad 0 \leq t \leq 1 \quad dx=-dt$

$$y=1-x^2 = 1-(1-t)^2 = \cancel{1} - \cancel{1} + 2t - t^2 \quad dy=2-2t$$

(II) der astern

$$\int_{C_3} x dx + y dy = \int_0^1 (1-t)(-dt) + (2t-t^2)(2-2t) dt$$

$$= \int_0^1 (t-1+4t-2t^2-4t^2+2t^3) dt$$

$$= 5 \cdot \frac{t^2}{2} - 6 \frac{t^3}{3} + 2 \frac{t^4}{4} - t \Big|_0^1 = 0$$

(11) Use Green's theorem to evaluate the line integral along the positively oriented curve

(a) $\int_C e^y dx + 2xe^y dy$ C : square with sides $x=0 \quad x=1 \quad y=0 \quad y=1$

(b) $\int_C (y + e^{\sqrt{x}}) dx + (2x + \cos y^2) dy$ C : boundary of the region closed by parabolas $y=x^2 \quad x=y^2$

(c) $\int_C x e^{-2x} dx + (x^4 + 2x^2 y^2) dy$ C : boundary of the region between circles

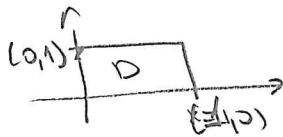
(d) $\int_C y^3 dx - x^3 dy$ C : circle $x^2 + y^2 = 4$ $x^2 + y^2 = 1 \quad x^2 + y^2 = 4$

(2)

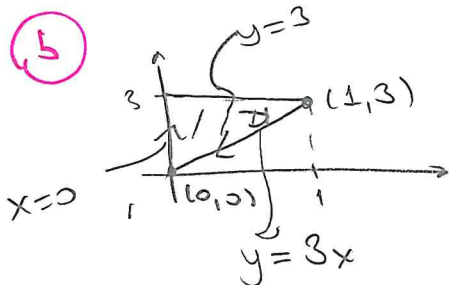
Deyan Atanasov

a) $\int_C e^y dx + 2xe^y dy = \iint_D (Q_x - P_y) dA = \int_0^1 \int_0^1 (2e^y - e^y) dx dy$
 $= \int_0^1 xe^y \Big|_0^1 dy$
 $= e - 1$

$Q_x = 2e^y$
 $P_y = e^y$



b) $\iint_D (Q_x - P_y) = \int_0^3 \int_0^{y/3} (4y^3 - 2x^2y) dx dy$
 $Q_x = 4y^3$
 $P_y = 2x^2y$

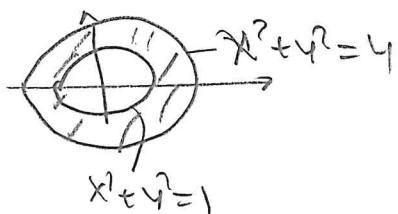


$= \int_0^3 \left(4y^3 \cdot x - 2 \frac{x^3}{3} \cdot y \right) \Big|_0^{y/3} dy = \int_0^3 \left(\frac{4}{3} y^4 - \frac{2}{81} y^4 \right) dy$
 $= \int_0^3 \left(\frac{108-2}{81} \right) y^4 dy = \frac{106}{81} \cdot \frac{y^5}{5} \Big|_0^3 = \frac{106}{81} \cdot \frac{3^5}{5} = \frac{316}{5}$

c) $\int_C \underbrace{x e^{-2x}}_P dx + \underbrace{(x^4 + 2x^2 y^2)}_Q dy = \iint_D (4x^3 + 4xy^2) dA = I$

$Q_x = 4x^3 + 4xy^2$
 $P_y = 0$

$Q_x - P_y = 4x^3 + 4xy^2 = 4x(x^2 + y^2)$



$x = r \cos \theta$
 $y = r \sin \theta$
 $x^2 + y^2 = r^2$
 $Q_x - P_y = 4r \cos \theta \cdot r^2$
 $I = \int_0^{2\pi} \int_0^2 4r^3 \cos \theta \cdot r dr d\theta$

$I = 4 \int_0^{2\pi} \frac{r^5}{5} \Big|_0^2 \cos \theta d\theta = 4 \left(\frac{2^5 - 1}{5} \right) \sin \theta \Big|_0^{2\pi} = 0$

Deeya Attintin

(d) $\int_P y^3 dx - x^3 dy = \int_0^{2\pi} \int_0^2 -3r^2 \cdot r dr d\theta = 3 \int_0^{2\pi} -\frac{r^4}{4} \Big|_0^2 d\theta = -24\pi$

$P = y^3 \quad P_y = 3y^2 \quad Q_x - P_y = -3x^2 - 3y^2 = -3(x^2 + y^2)$
 $Q = -x^3 \quad Q_x = -3x^2 \quad Q_x - P_y = -3r^2$

$C: x^2 + y^2 = 4 \quad x = r \cos \theta \quad 0 \leq r \leq 2 \quad x^2 + y^2 = r^2$
 $y = r \sin \theta \quad 0 \leq \theta \leq 2\pi$

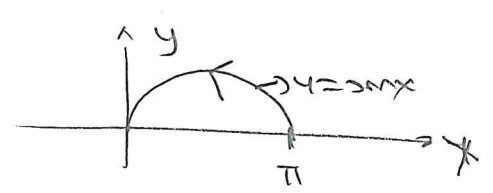
(12) Use Green's Theorem to evaluate $\int_C F \cdot dr$ (Check orientation of C)

(a) $F(x,y) = \langle \sqrt{x} + y^3, x^2 + \sqrt{y} \rangle$ C: $y = \sin x$ from $(0,0)$ to $(\pi,0)$
 and line segment from $(\pi,0)$ to $(0,0)$

(b) $F(x,y) = \langle y^2 \cos x, x^2 + 2y \sin x \rangle$ C: triangle from $(0,0)$ to $(2,6)$
 to $(2,0)$ to $(0,0)$

(c) $F(x,y) = \langle \sqrt{x} + y^3, x^2 + \sqrt{y} \rangle$
 $dr = \langle dx, dy \rangle$ Please be careful C is clockwise
 So, $-C$ gives positive orientation counter clockwise

$\int_C F \cdot dr = - \int_{-C} F \cdot dr = - \int_{-C} (\underbrace{\sqrt{x} + y^3}_P) dx + (\underbrace{x^2 + \sqrt{y}}_Q) dy = - \int_0^\pi \int_0^{\sin x} (2x - 3y^2) dy dx$



$Q_x = 2x \quad (Q_x - P_y) = 2x - 3y^2$
 $P_y = 3y^2$

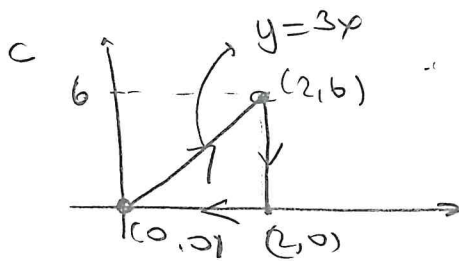
$= - \int_0^\pi (2xy - y^3) dx \Big|_0^{\sin x} = - \int_0^\pi (2x \sin x - (\sin x)^3) dx$
 $= - [2x \sin x - 2x \cos x + \cos x - \cos^3 x] \Big|_0^\pi = \frac{4}{3} - 2\pi$

$\int x \sin x dx = -x \cos x + \sin x$
 $x = u \quad \sin x dx = du$

(13)

Derja Altinter

(b)



C is in the negative direction clockwise. To apply Green's formula we must take $-C$.

$$\int_C F \cdot dr = - \int_{-C} F \cdot dr = - \int_{-C} \underbrace{y^2 \cos x}_{P} dx + \underbrace{(x^2 + 2y \sin x)}_Q dx$$

$$Q_x = 2x + 2y \cos x \quad Q_x - P_y = 2x$$

$$P_y = 2y \cos x$$

$$= - \int_0^2 \int_0^{3x} 2x \, dy \, dx = - \int_0^2 2x \cdot 3x \, dx = - 2x^3 \Big|_0^2 = -16$$

Deyan Altiman

(14)

13 Find (a) Curl (b) Divergence of the following vector fields.

(a) $F(x, y, z) = x^2 y z \vec{i} + x y^2 z \vec{j} + x y z^2 \vec{k}$

(b) $F(x, y, z) = \vec{i} + (x + y z) \vec{j} + (x y - \sqrt{z}) \vec{k}$

(c) $F(x, y, z) = \cos x z \vec{j} - \sin x y \vec{k}$

(a)
$$\text{Curl } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y z & x y^2 z & x y z^2 \end{vmatrix} = (x z^2 - x y^2) \vec{i} - (y z^2 - x^2 z) \vec{j} + (y^2 z - x^2 z) \vec{k}$$

$$= x(z^2 - y^2) \vec{i} + y(x^2 - z^2) \vec{j} + z(y^2 - x^2) \vec{k}$$

$$\text{div } F = \nabla \cdot F = \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (x y^2 z) + \frac{\partial}{\partial z} (x y z^2) = 2x y z + 2x y z + 2x y z = 6x y z$$

(b)
$$\text{Curl } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & x + y z & x y - \sqrt{z} \end{vmatrix} = (x - y) \vec{i} - (y - 0) \vec{j} + (1 - 0) \vec{k}$$

$$= (x - y) \vec{i} - y \vec{j} + \vec{k}$$

$$\text{div } F = \nabla \cdot F = \frac{\partial}{\partial x} (1) + \frac{\partial}{\partial y} (x + y z) + \frac{\partial}{\partial z} (x y - \sqrt{z})$$

$$= z - \frac{1}{2\sqrt{z}}$$

(c)
$$\text{Curl } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \cos x z & -\sin x y \end{vmatrix}$$

$$= (-x \cos x y + \cos x z) \vec{i} - (-y \cos x y - 0) \vec{j} + (-z \sin x z - 0) \vec{k}$$

$$= x(\sin x z - \cos x y) \vec{i} + y \cos x y \vec{j} - z \sin x z \vec{k}$$

$$\text{div } F = \nabla \cdot F = \frac{\partial}{\partial x} (0) + \frac{\partial}{\partial y} (\cos x z) + \frac{\partial}{\partial z} (-\sin x y) = 0$$

(15)

Dey's Attention

14) Determine whether or not the vector field is conservative.

If it is conservative, find a function f such that $F = \nabla f$

a) $F(x, y, z) = yz\vec{i} + xz\vec{j} + xy\vec{k}$

b) $F(x, y, z) = 3z^2\vec{i} + \cos y\vec{j} + 2xz\vec{k}$

c) $F(x, y, z) = e^{xz}\vec{i} + \frac{z}{y}\vec{j} + xe^{xz}\vec{k}$

a) Check $\text{curl } F = \nabla \times F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix} = \vec{i}(x-x) - \vec{j}(y-y) + \vec{k}(z-z) = \vec{0}$

Since $\text{curl } F = \vec{0}$. Since F is defined on all $\mathbb{R}^3$ by
Then it is conservative.
Find f such that $F = \nabla f$

$$f_x = yz \quad f_y = xz \quad f_z = xy$$

$$\Downarrow$$

$$f = xyz + h(y, z)$$

$$f_y = xz + h'(y, z) = xz$$

$$h'(y, z) = 0$$

$$h(y, z) = g(z) \quad f(x, y, z) = xyz + g(z)$$

$$f_z = xy + g'(z) = xy \quad g'(z) = 0 \quad g(z) = C$$

$$f(x, y, z) = xyz + C$$

b) $\text{curl } F = \nabla \times F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3z^2 & \cos y & 2xz \end{vmatrix}$

$$= \vec{i}(0-0) - \vec{j}(2z-6z) + (0-0)\vec{k} = 4z\vec{j} \neq \vec{0}$$

F is not conservative

Deya Altıntaş

$$\textcircled{c} \text{Curl } F = \nabla \times F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^z & 1 & xe^z \end{vmatrix}$$

$$= (0-0)\vec{i} - (e^z - e^z)\vec{j} + (0-0)\vec{k} = 0$$

F defined in $\mathbb{R}^3$ and $\text{Curl } F = 0$ so it is conservative.

$$F = \nabla f = \langle f_x, f_y, f_z \rangle = \langle e^z, 1, xe^z \rangle$$

$$f_x = e^z \quad f_y = 1 \quad f_z = xe^z$$

$\Downarrow$

$$f(x, y, z) = xe^z + h(y, z) \quad f_y(x, y, z) = h'(y, z) = 1$$

$$h(y, z) = y + g(z) \quad f(x, y, z) = xe^z + y + g(z)$$

$$f_z = x \cdot e^z + g'(z) = xe^z \quad g'(z) = 0 \quad g(z) = C$$

$$f(x, y, z) = xe^z + y + C$$

$\textcircled{15}$ Is there a vector field G on $\mathbb{R}^3$ such that $\text{Curl } G = yz\vec{i} + xy\vec{j} + xz\vec{k}$ Explain

If it were true such that $F = \text{Curl } G$, then

we must have $\text{div } F = \text{div}(\text{Curl } G) = 0$

$$\text{div}(\text{Curl } G) = \underbrace{\frac{\partial}{\partial x}(yz)}_0 + \underbrace{\frac{\partial}{\partial y}(xy)}_x + \underbrace{\frac{\partial}{\partial z}(xz)}_x = xz \neq 0$$

Therefore, $\text{Curl } G$ can't be a vector field.

$\textcircled{16}$ Show that any vector field of the form

$$F(x, y, z) = f(y, z)\vec{i} + g(x, z)\vec{j} + h(x, y)\vec{k}$$

is incompressible.

Deyan Atkinson

$\textcircled{17}$

$$\text{div } F = \frac{\partial (f(x,y,z))}{\partial x} + \frac{\partial (g(x,y,z))}{\partial y} + \frac{\partial (h(x,y,z))}{\partial z} = 0 \text{ so } F \text{ is}$$

incompressible.

17 Show that any vector field of the form

$$F(x,y,z) = f(x)\vec{i} + g(y)\vec{j} + h(z)\vec{k}$$

where f, g, h are differentiable functions, is irrotational.

$$\text{curl } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f(x) & g(y) & h(z) \end{vmatrix} = \vec{i}(0-0) + (0-0)\vec{j} + \vec{k}(0-0) = 0$$

Hence, F is irrotational.

18 Use Stokes theorem's to evaluate $\iint_S \text{curl } F \cdot d\vec{S}$

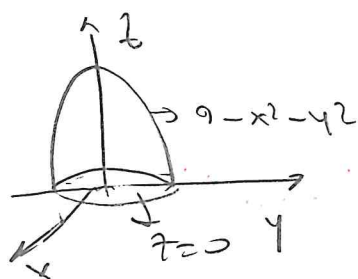
a) $F(x,y,z) = yz\vec{i} + xz\vec{j} + xy\vec{k}$ S is the part of the paraboloid

$z = 9 - x^2 - y^2$ that lies above the plane $z = 5$, oriented upward

b) $F(x,y,z) = x^2 e^{yz}\vec{i} + y^2 e^{xz}\vec{j} + z^2 e^{xy}\vec{k}$ S is the hemisphere $x^2 + y^2 + z^2 = 4$ $z > 0$ oriented upward.

y

c) $z = 9 - x^2 - y^2 \wedge z = 5 \quad 9 - x^2 - y^2 = 5 \quad x^2 + y^2 = 4$



Boundary C is oriented in the counterclockwise

direction so $r(t) = 2\cos t\vec{i} + 2\sin t\vec{j} + 5\vec{k}$

$$0 \leq t \leq 2\pi \quad r'(t) = -2\sin t\vec{i} + 2\cos t\vec{j} + 0\vec{k}$$

Daya Altıntaş

18

$$F(r(t)) = 10 \sin t \vec{i} + 10 \cos t \vec{j} + 4 \cos t \sin t \vec{k} \quad \text{by Stokes thm}$$

$$\begin{aligned} \iint_S \text{curl } F \cdot d\vec{s} &= \int_C F \cdot d\vec{r} = \int_0^{2\pi} F(r(t)) \cdot r'(t) dt = \int_0^{2\pi} (-20 \sin^2 t + 20 \cos^2 t) dt \\ &= 20 \int_0^{2\pi} \cos 2t dt = 0 \end{aligned}$$

5) $x^2 + y^2 + z^2 = 4 \wedge z \geq 0$ we have $x^2 + y^2 = 4 \quad 0 \leq \theta \leq 2\pi$
 $r(t) = 2 \cos t \vec{i} + 2 \sin t \vec{j} + 0 \vec{k} \quad r'(t) = -2 \sin t \vec{i} + 2 \cos t \vec{j}$
 $F(r(t)) = 4 \cos^2 t e^{\frac{1}{2}} \vec{i} + 4 \sin^2 t e^{\frac{1}{2}} \vec{j} + 0 \vec{k}$
 $= 4 \cos^2 t \vec{i} + 4 \sin^2 t \vec{j} + 0 \vec{k}$

$$\begin{aligned} \iint_S \text{curl } F \cdot d\vec{s} &= \int_C F \cdot d\vec{r} = \int_0^{2\pi} (-8 \cos^2 t \sin t + 8 \sin^2 t \cos t) dt \\ &= -8 \cdot \left(\frac{1}{3} \cos^3 t + \frac{1}{3} \sin^3 t \right) \Big|_0^{2\pi} = 0 \end{aligned}$$

19) Use Stokes Thm to evaluate $\int_C F \cdot d\vec{r}$. In each case C is oriented counterclockwise as viewed from above.

a) $F(x, y, z) = e^{-x} \vec{i} + e^x \vec{j} + e^z \vec{k} \quad C: \text{Boundary of the part of the plane } 2x + y + 2z = 2 \text{ in the first octant}$

b) $F(x, y, z) = y \vec{i} + 2xz \vec{j} + e^{xy} \vec{k} \quad C: x^2 + y^2 = 16, z = 5$

a) $\text{curl } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{-x} & e^x & e^z \end{vmatrix} = \vec{i}(0) - \vec{j}(0) + \vec{k}(e^x) = \langle 0, 0, e^x \rangle$

Deyu Altun

$$z = g(x, y) = 1 - x - y/2$$

$$\nabla f = \langle 1, 1/2, 1 \rangle$$

$f(x, y, z) = z - 1 + x + y/2$ Use convert it this form because of the denominator of surface integral

19

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D \text{curl } \mathbf{F} \cdot \vec{\nabla} f \cdot dA =$$

$$= \int_0^1 \int_0^{2-2x} e^x dy dx = \int_0^1 (2-2x)e^x dx = (2-2x)e^x + 2e^x \Big|_0^1 = (2e-4)$$

$$2x+4+2z=2. \quad \begin{matrix} z=0 \\ \Rightarrow \end{matrix} \quad D: 2x+4=2$$

$$\int x e^x dx = x \cdot e^x - \int e^x dx = x e^x - e^x$$

$$x=u \quad e^x dx = du$$

$$dx=du \quad e^x=u$$

⑤ $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$

$$C: \text{ is the circle } x^2 + y^2 = 16 \quad z=5.$$

$$f(x,y,z) = z-5 \quad \nabla f = \langle 0, 0, 1 \rangle$$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & 2xz & e^{xy} \end{vmatrix} = \vec{i} (x e^{xy} - 2x) - \vec{j} (y e^{xy} - y) + \vec{k} (2z - z)$$

$$= \langle x e^{xy} - 2x, y - y e^{xy}, 2z - z \rangle$$

$$\text{curl } \mathbf{F} \cdot \nabla f = 2z - z$$

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^4 (2 \cdot 5 - 5) r dr d\theta = 5 \cdot (4^2 \pi) = 80\pi$$

If we try to solve it from $\int \mathbf{F} \cdot d\mathbf{r}$

$$\mathbf{r}(t) = 4 \cos t \vec{i} + 4 \sin t \vec{j} + 5 \vec{k} \quad 0 \leq t \leq 2\pi \quad \mathbf{r}'(t) = -4 \sin t \vec{i} + 4 \cos t \vec{j} + 0 \vec{k}$$

$$\mathbf{F}(\mathbf{r}(t)) = 20 \sin t \vec{i} + 40 \cos t \vec{j} + e^{16 \sin t \cos t} \vec{k} \quad \text{Degree Attention}$$

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -80 \sin^2 t + 160 \cos^2 t = 80(\cos^2 t - \sin^2 t) + 80 \cos^2 t$$

$$\cos^2 t = \frac{\cos 2t + 1}{2} \quad = 120 \cos 2t + 40 \quad \cos^2 t = \frac{\cos 2t + 1}{2}$$

$$\int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = 120 \frac{\sin 2t}{2} \Big|_0^{2\pi} + 40 \Big|_0^{2\pi} = 80\pi$$